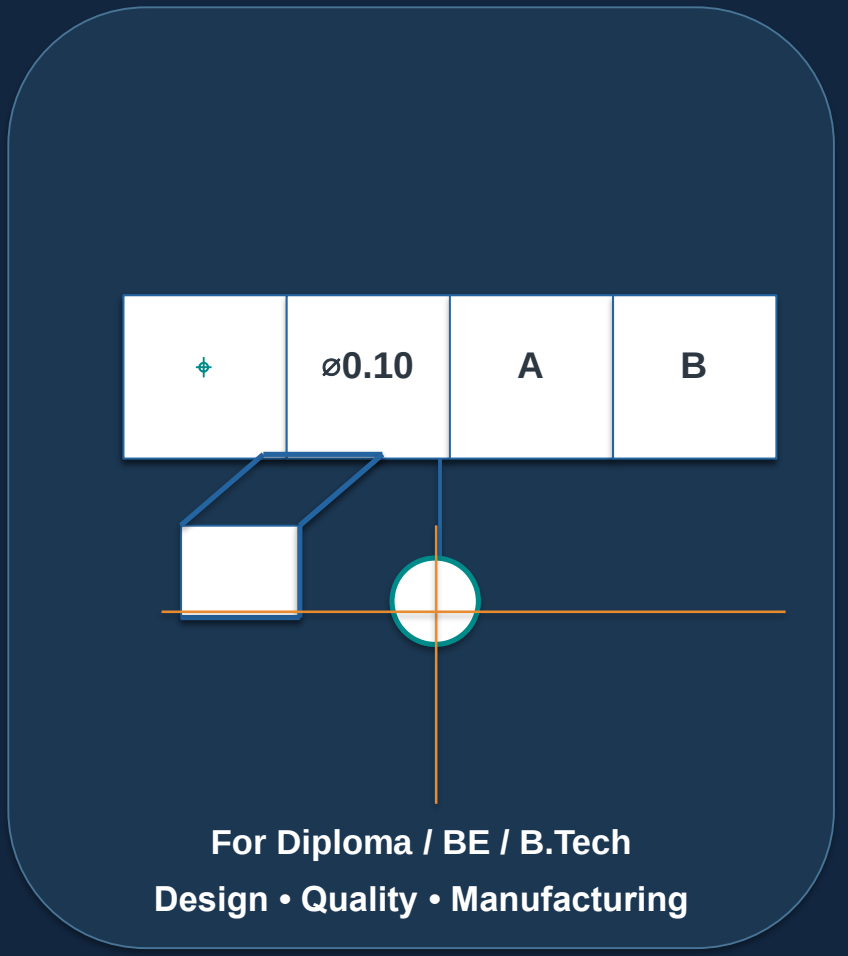
GD&T

GEOMETRIC DIMENSIONING & TOLERANCING

Complete Mechanical Engineering Guide

Symbols • Datums • Feature Control Frames • Tolerance Zones • Inspection



For Diploma / BE / B.Tech

Design • Quality • Manufacturing

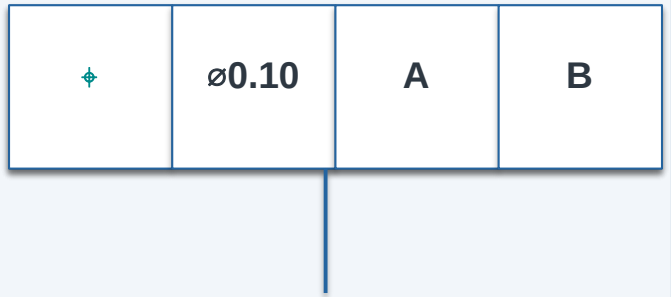
Illustrations are schematic. Always follow the drawing standard and revision specified by your organization.

1. Learning Objectives

What you will be able to do after this presentation

- Understand why geometric tolerances are used beyond size tolerances.
- Read feature control frames (FCFs), datum references and common symb
- Distinguish form, orientation, location, profile and runout controls.
- Understand tolerance zones, MMC/LMC/RFS concepts and bonus toleran
- Plan inspection using gauges, indicators, CMMs and functional fixtures.

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KEY POINT

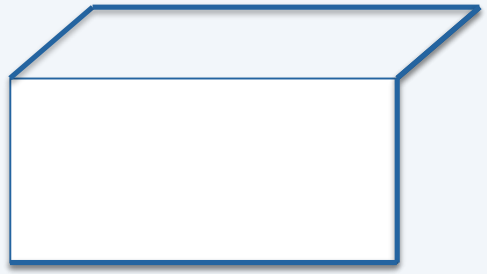
Read the control, identify the datum, understand the zone.

2. What Is GD&T?

Definition and purpose

- GD&T is a symbolic language used on engineering drawings to define allowa
- It controls form, orientation, location, profile and runout of features.
- It communicates design intent more clearly than coordinate dimensions alo
- GD&T helps ensure assembly, interchangeability, function and consistent i
- Common standards include ASME Y14.5 and ISO GPS standards; require

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KEY POINT

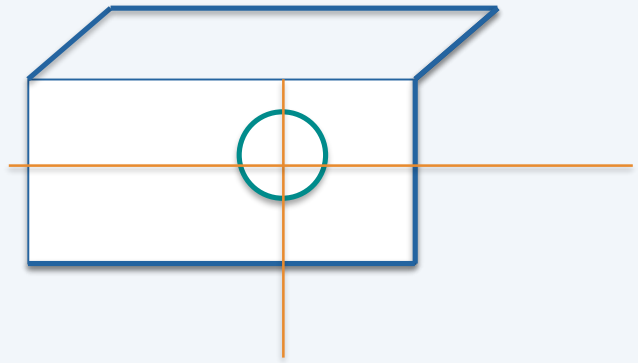
GD&T controls geometry, not just feature size.

3. Why Size Tolerances Are Not Enough

Size vs. geometry

- A hole can be within diameter limits but still be in the wrong position.
- A surface may meet a size requirement yet be bowed, twisted or tilted.
- GD&T defines how a feature relates to ideal geometry and functional datu
- It can permit useful variation while protecting assembly function.
- Example: a bolt pattern needs both hole-size limits and positional control.

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KEY POINT

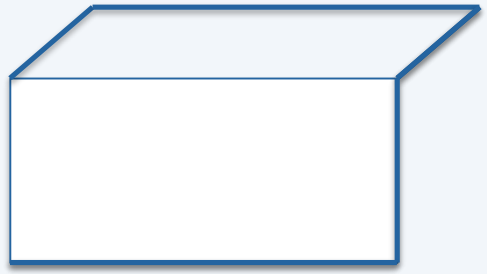
Size compliance does not guarantee location or orientation.

4. Basic Drawing Language

Terms used throughout GD&T

- Feature: a physical point, line, surface, hole, slot or derived axis/plane.
- Nominal/true geometry: the theoretically exact shape or location.
- Actual geometry: the manufactured feature, which varies within permitted l
- Tolerance zone: the region within which the controlled feature must lie.
- Basic dimensions: theoretically exact dimensions, commonly boxed on dra

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KEY POINT

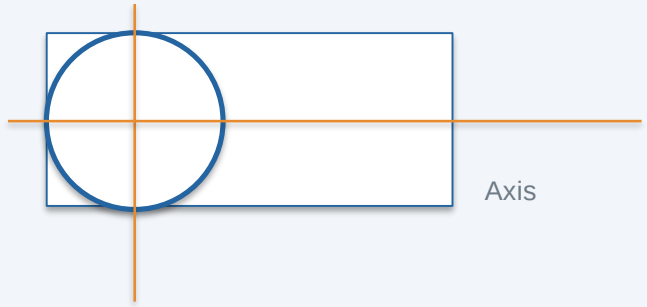
Basic dimensions locate the ideal zone; the geometric toleranc

5. GD&T Control Families

Five commonly taught control families

- Form: straightness, flatness, circularity and cylindricity.
- Orientation: parallelism, perpendicularity and angularity.
- Location: position; some standards also define concentricity and symmetry
- Profile: profile of a line and profile of a surface.
- Runout: circular runout and total runout.

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KEY POINT

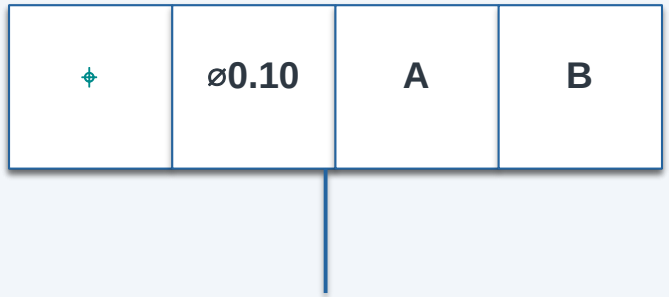
Choose the control based on functional intent.

6. How to Read a Feature Control Frame

Feature Control Frame (FCF)

- First compartment: geometric characteristic symbol.
- Next compartment: tolerance value, sometimes preceded by diameter sym
- Material-condition modifier may follow the tolerance value: Ⓜ MMC or Ⓛ L er ASM
- Datum compartments identify the referenced datum features in order.
- Read the complete frame together with the feature, dimensions and notes.

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KEY POINT

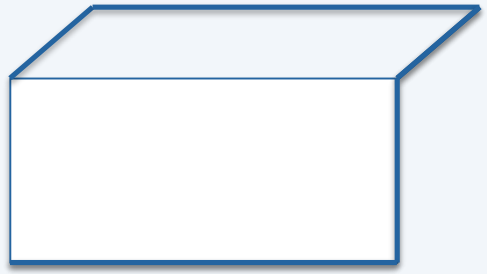
Symbol + tolerance zone + modifiers + datum references.

7. Datums and Datum Features

Reference framework

- A datum is a theoretically exact reference derived from a real datum feature.
- A datum feature is the physical feature identified on the part, such as a fac
- Datum simulators represent ideal planes, pins or other perfect references
- Datum symbols are attached to the feature outline, extension line or dimer
- Datum references constrain the part in a repeatable coordinate system.

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A

KEY POINT

A datum is ideal; a datum feature is real.

8. 3-2-1 Datum Setup Concept

Conceptual datum reference frame

- Primary datum typically constrains three degrees of freedom.
- Secondary datum typically constrains two additional degrees of freedom.
- Tertiary datum constrains the final remaining degree of freedom.
- Exact constraints depend on feature geometry and datum simulator contac
- Use a stable, functional setup that reproduces how the part assembles or

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KEY POINT

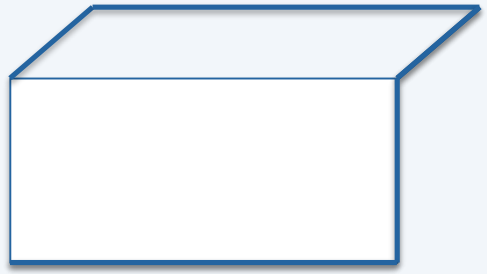
A 3-2-1 setup is a useful model, not a substitute for standard-sp

9. Datum Order: A | B | C

Datum reference sequence

- Datum A is the primary reference; B is secondary; C is tertiary in the referenc
- Order is meaningful: changing the order can change how the tolerance is e
- Primary datum establishes the first and most important orientation/constra
- Datum features should be selected from functional interfaces where possib
- Do not assume every drawing needs three datums; use only those require

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KEY POINT

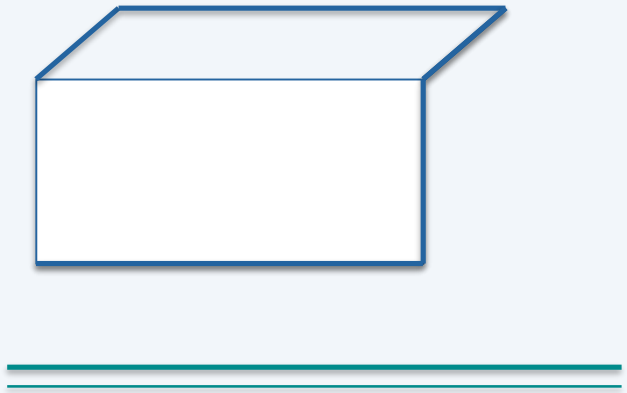
Use functional datum selection—not simply the easiest surface

10. Straightness

Form control

- Controls how straight a line element or derived median line must be.
- Surface-line straightness: each considered line element must lie between t
- Axis/median-line straightness can use a cylindrical tolerance zone when th
- Straightness is a form control and does not require a datum reference.
- Check the exact application and symbol rules in the governing standard.

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KEY POINT

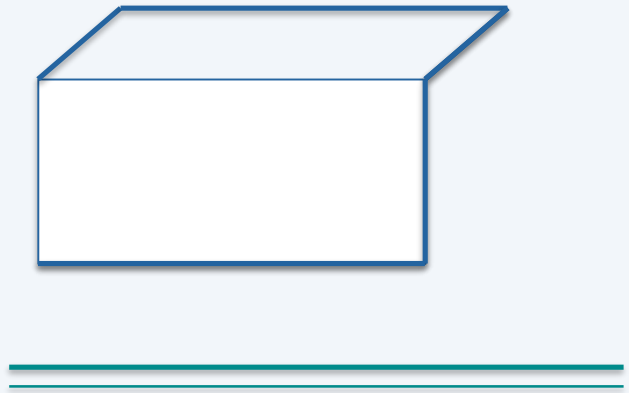
No datum is required for a form tolerance.

11. Flatness

Form control

- Controls all points of a surface to lie between two parallel planes.
- Flatness does not require a datum reference.
- Tolerance value is the separation between the two planes.
- Surface finish, fixture distortion and measurement strategy can affect resu
- Inspection examples: surface plate with indicator, CMM, or suitable flatnes

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KEY POINT

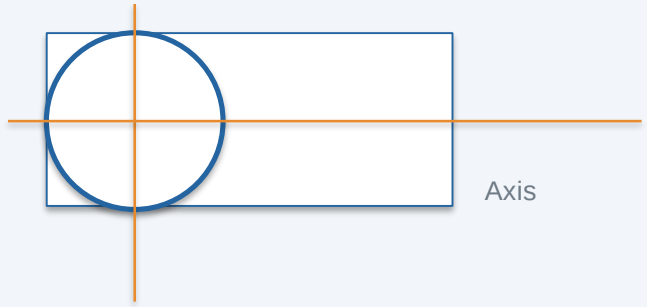
Flatness controls the surface itself, not its orientation to a datu

12. Circularity (Roundness)

Form control

- Controls each circular cross-section so its points lie between two concentric ci
- It does not control the full cylinder's straightness or taper.
- It does not require a datum reference.
- Often checked with a roundness tester or suitable rotary measurement sys
- Do not confuse circularity with circular runout; runout references a datum a

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KEY POINT

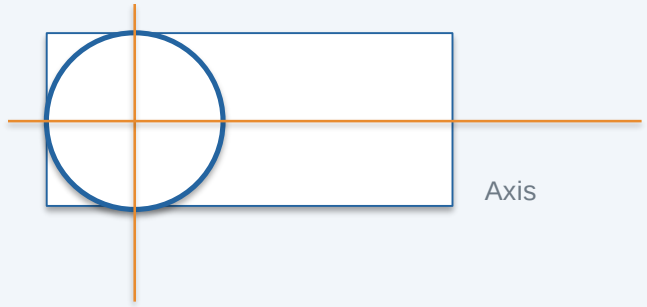
Circularity is evaluated in individual cross-sections.

13. Cylindricity

Form control

- Controls the entire cylindrical surface between two coaxial cylinders.
- It combines requirements related to circularity, straightness and taper over
- It is a form control and does not require a datum reference.
- Measurement may require a CMM or specialized form-measuring equipme
- Use it when the entire cylindrical form is functionally important.

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KEY POINT

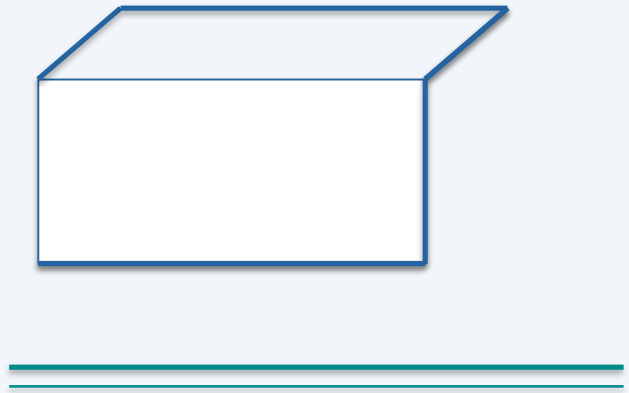
Cylindricity controls the whole surface, not just one section.

14. Parallelism

Orientation control

- Controls a surface, line element or derived axis to be parallel to a datum.
- The tolerance zone may be two parallel planes or a cylindrical zone, deper sage.
- Datum reference is required.
- Parallelism does not by itself control the feature's location.
- Example: a guide rail surface kept parallel to a base datum.

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KEY POINT

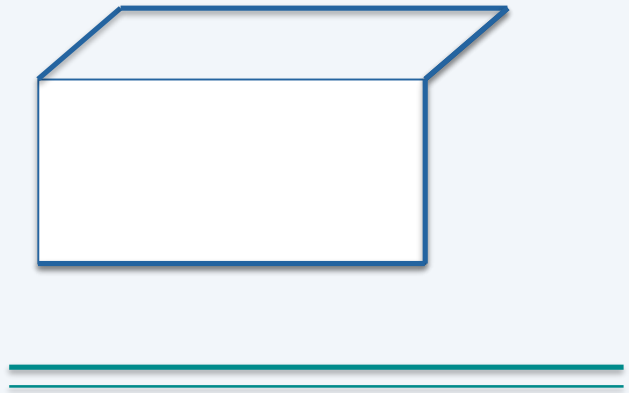
Parallelism controls orientation relative to a datum.

15. Perpendicularity

Orientation control

- Controls a surface, axis or center plane at 90° to a datum.
- Tolerance zone depends on whether a surface, axis or center plane is cont
- Datum reference is required.
- Basic dimensions may define the theoretically exact 90° relationship.
- Example: a mounting face perpendicular to a locating base.

SCHEMATIC



KEY POINT

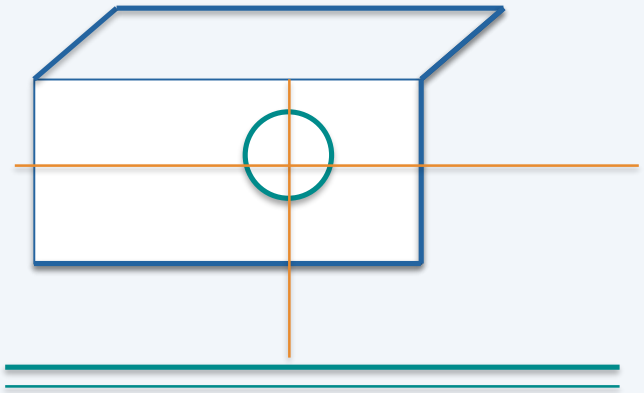
Perpendicularity controls 90° orientation.

16. Angularity

Orientation control

- Controls a surface, axis or center plane at a specified basic angle to a datum
- The angle is theoretically exact; the tolerance zone permits variation aroun
- Datum reference is required.
- Use basic angle dimensions with the appropriate geometric tolerance.
- Example: a machined ramp held at 30° relative to a base.

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KEY POINT

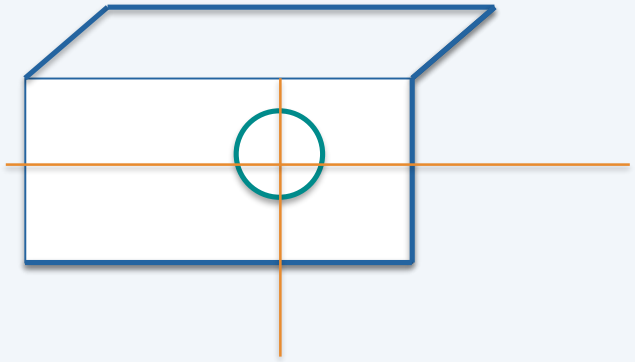
Angularity is for an angle other than the 90° controlled by perp

17. Position (True Position)

Location control

- Controls the location of a feature of size, such as a hole or pin, relative to dat
- Often used with basic dimensions to locate hole centers or axes.
- A diametrical tolerance zone is common for axes of holes/pins.
- Datum references establish the coordinate frame; modifiers may affect the
- Position is widely used for bolt patterns and interchangeable assemblies.

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KEY POINT

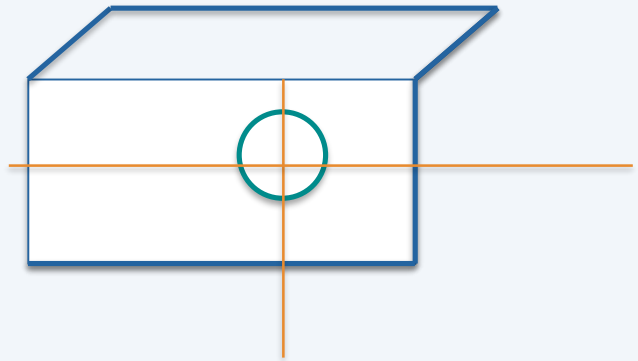
Position is not the same as measuring X and Y deviations inde

18. Position Tolerance Zone

Understanding the zone

- For a hole axis, a diameter symbol before the tolerance commonly indicates a
- The actual axis/derived median line must remain within the specified zone.
- Basic dimensions locate the zone theoretically exactly.
- Datum mobility and material-condition modifiers can change how the zone
- Always interpret the feature control frame, feature size and datum modifier

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KEY POINT

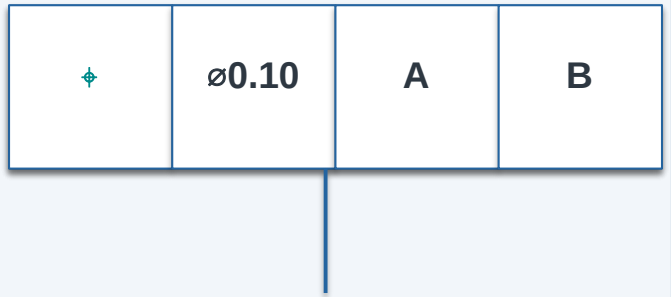
A ⌀ tolerance defines a cylindrical zone when the standard and

19. MMC, LMC and RFS

Material condition concepts

- MMC: maximum material condition—most material in the feature; smallest ho
- LMC: least material condition—least material; largest hole or smallest shaf
- RFS: regardless of feature size; tolerance applies independently of actual
- Material-condition modifiers are applied only where permitted by the gover
- ISO GPS uses its own rules and terminology; do not mix systems without

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KEY POINT

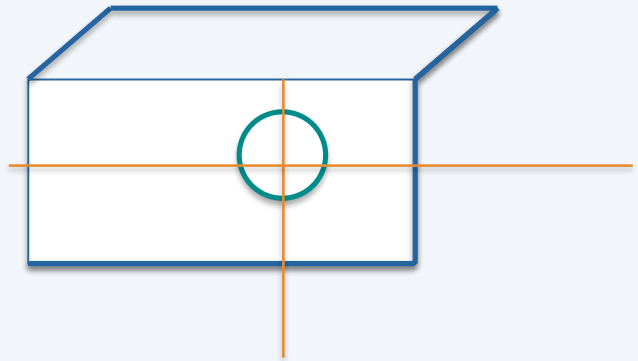
For a hole: MMC is its smallest permitted diameter.

20. Bonus Tolerance and Virtual Condition

Calculation concept

- When a position tolerance is specified at MMC, bonus tolerance is generally a …C.
- Hole example: bonus = actual hole size − hole MMC size.
- Virtual condition for an internal feature under common ASME MMC applica… e.
- External features use the opposite size direction for virtual-condition calcul…
- These rules depend on feature type, datum modifiers and the standard; ve…

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KEY POINT

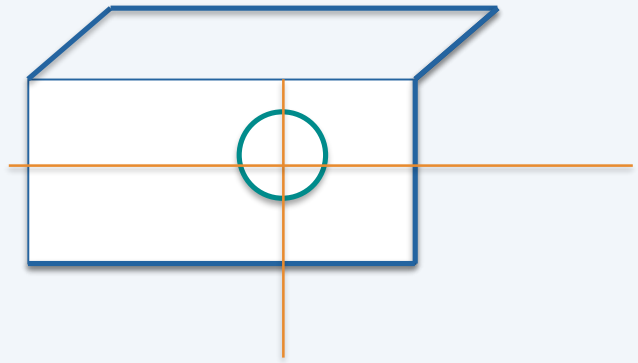
Bonus tolerance is not extra size tolerance; it is extra geometri

21. Worked Example: Hole Position

Worked calculation

- Drawing: hole diameter 10.0–10.2 mm; position ⌀0.2 at MMC relative to A|B|C
- Hole MMC = 10.0 mm. Actual measured hole = 10.1 mm.
- Bonus tolerance = 10.1 − 10.0 = 0.1 mm.
- Total allowed position tolerance = 0.2 + 0.1 = ⌀0.3 mm.
- Assumes standard-compatible MMC application and no additional datum-s

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KEY POINT

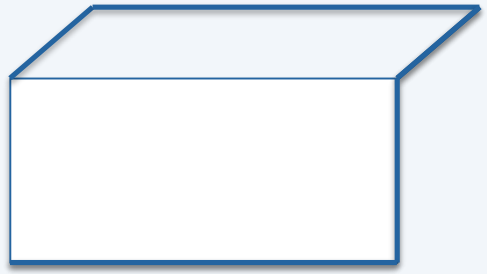
Actual hole size increases clearance, so allowable position tole

22. Profile of a Line

Profile control

- Controls the shape of a line element of a surface in a specified section.
- Tolerance zone is usually two offset curves around the true profile.
- May be applied with or without datum references depending on design inte
- Can control complex contours that are difficult to dimension with many coo
- Specify the section/extent clearly when needed.

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KEY POINT

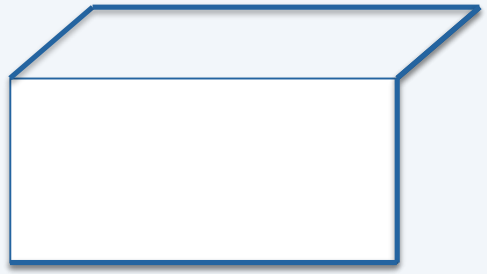
Profile can control complex geometry with one coherent requir

23. Profile of a Surface

Profile control

- Controls the entire surface relative to a true profile.
- Tolerance zone is typically two offset surfaces surrounding the theoretically
- Can control form, orientation and location depending on datum references
- Often used for castings, moulded parts, airfoils, ergonomic surfaces and c
- Inspect using CMM, scanning, templates or dedicated fixtures as appropri

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KEY POINT

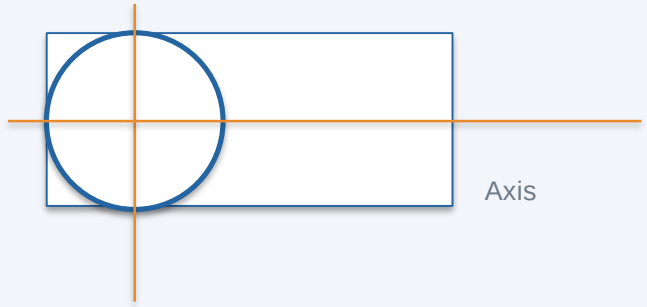
Datum references determine how the profile is located and orie

24. Circular Runout

Runout control

- Controls variation of a surface during one full rotation about a datum axis.
- Measured at each circular cross-section while the part rotates.
- Combines effects of surface form and eccentricity relative to the datum axi
- Requires a datum axis, typically established from a bearing journal or othe
- Often inspected with a dial indicator and suitable rotary setup.

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KEY POINT

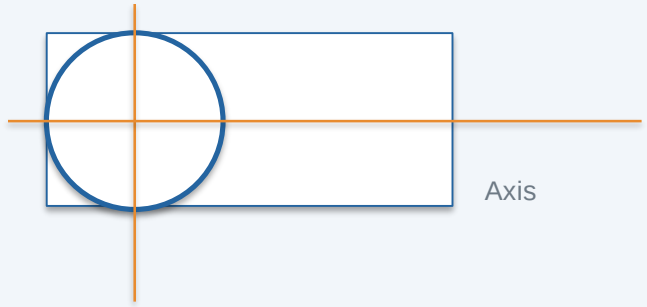
Circular runout is checked section by section.

25. Total Runout

Runout control

- Controls variation over the full surface while rotating the part about a datum a
- Indicator traverses along the surface as the part rotates.
- Controls cumulative effects including form, alignment and surface variation
- Requires a datum axis and a suitable measurement setup.
- Common for shafts, spindles, sealing surfaces and rotating components.

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KEY POINT

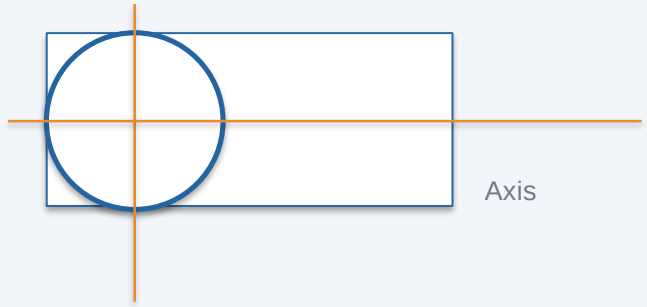
Total runout evaluates the whole controlled surface.

26. Concentricity and Symmetry: Caution

Legacy drawing awareness

- These controls are found in some editions/standards and may be treated diffe
- ASME Y14.5-2018 removed concentricity and symmetry as geometric cont
- Many functional applications can be better controlled with position, runout
- Do not silently reinterpret legacy symbols—confirm the drawing standard a
- ISO GPS terminology and rules are not identical to ASME.

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KEY POINT

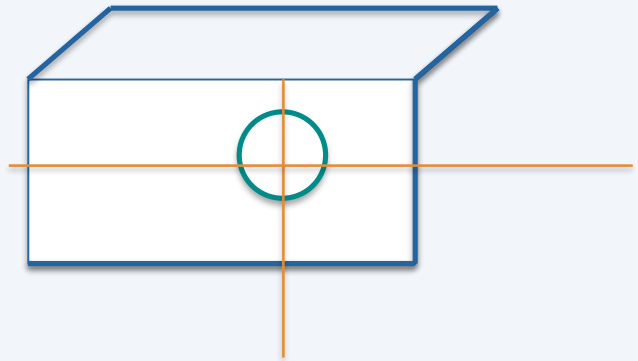
Always identify the applicable standard and edition.

27. Projected Tolerance Zone

Special application

- A projected tolerance zone extends beyond the part surface by a specified he
- Useful when a fastener or pin must remain perpendicular over its projectin
- Often applied to position of holes for studs, dowels or threaded fasteners.
- Projection height is specified with the applicable symbol and value.
- Inspection should represent the functional projecting element.

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KEY POINT

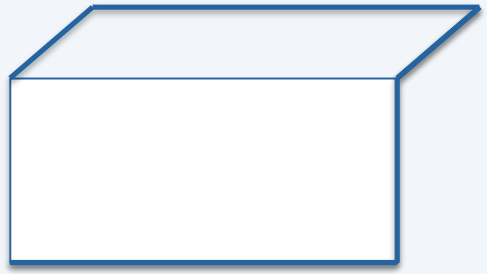
Use when alignment matters beyond the surface of the part.

28. Tangent Plane and Unequally Disposed Profile

Advanced interpretation

- Tangent-plane modifier can define a perfect plane tangent to high points of a
- Unequally disposed profile allocates different portions of a profile tolerance
- These controls address specific functional requirements and are standard-
- Do not assume a profile zone is equally split if a modifier says otherwise.
- Read all modifiers and notes before inspection.

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KEY POINT

Modifiers change how the tolerance zone is positioned or evalu

29. Free State and Flexible Parts

Special conditions

- Flexible, non-rigid parts may deform under their own weight or inspection forc
- Free-state requirements specify evaluation without externally applied restra
- Restrained-condition requirements define a fixture or loading state for insp
- Drawings must clearly state the applicable condition and any force/fixture c
- Examples include thin sheet metal, rubber seals and plastic covers.

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KEY POINT

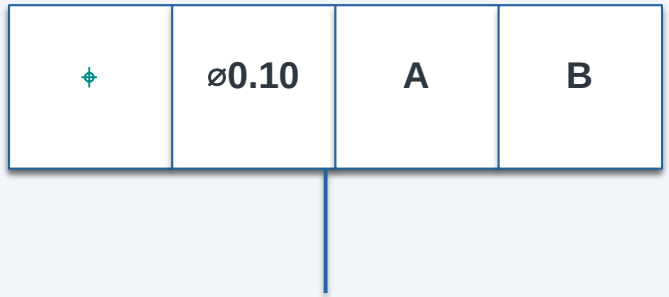
Inspect flexible parts in the condition specified by the drawing.

30. Common GD&T Symbols: Quick Reference

Symbol overview

- Straightness — line; flatness — parallelogram; circularity — circle; cylindricity
- Profile of line — arc/profile symbol; profile of surface — profile symbol with
- Parallelism — parallel lines; perpendicularity — right-angle symbol; angula
- Position — target/crosshair; circular runout — single runout arrow; total ru
- Symbol appearance can vary by font/rendering; use the official symbol tab

SCHEMATIC



KEY POINT

Verify exact symbols in the drawing standard; fonts may rende

31. GD&T Inspection Equipment

Measurement methods

- Calipers and micrometers: verify feature size, not all geometric controls.
- Dial indicators and fixtures: useful for runout, alignment and comparative c
- Surface plate, height gauge and squares: support basic setup and orientat
- CMM/vision/scanning systems: evaluate complex geometry and datum rel
- Functional gauges: fast acceptance checks when designed to the correct v

SCHEMATIC



KEY POINT

Use equipment suited to the tolerance and measurement uncer

32. Inspection Planning Checklist

Practical workflow

- Confirm drawing revision, units and governing GD&T standard/edition.
- Identify the controlled feature, FCF, basic dimensions, datum features and
- Choose a datum setup that simulates functional assembly and is repeatab
- Select equipment with adequate resolution and measurement uncertainty.
- Define sampling, temperature, cleanliness, fixturing and reporting requirem
- Document results and resolve ambiguous or conflicting requirements with

SCHEMATIC



KEY POINT

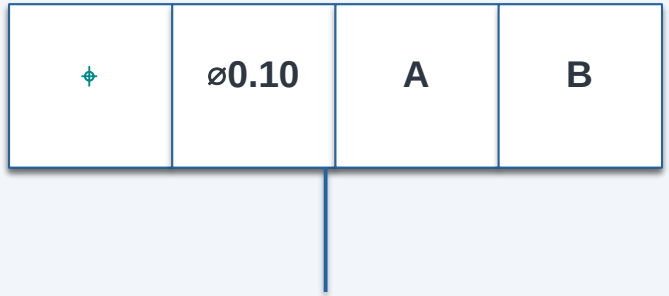
Never inspect from a cropped drawing without the notes and st

33. Common GD&T Mistakes

Avoid these errors

- Confusing datum and datum feature.
- Treating basic dimensions as plus/minus tolerances.
- Reading a diametrical position tolerance as separate X and Y limits.
- Ignoring MMC/LMC modifiers or incorrectly calculating bonus tolerance.
- Using runout when a form control alone is intended—or vice versa.
- Mixing ASME and ISO rules without checking the specified standard.

SCHEMATIC



KEY POINT

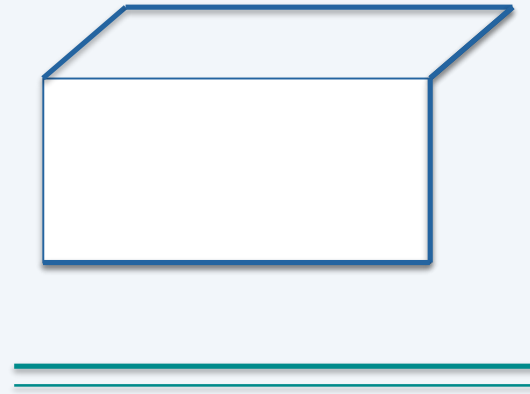
The full drawing specification governs acceptance.

34. Worked Example: Datum-Based Perpendicularity

Inspection example

- Requirement: a mounting face must be perpendicular to datum A within 0.05
- Set datum A on an appropriate simulator or surface plate, per the drawing
- Measure the controlled face with a height gauge/indicator or CMM.
- Evaluate the full surface against the permitted pair of parallel planes orient
- Do not judge perpendicularity only from one edge or one point.

SCHEMATIC



KEY POINT

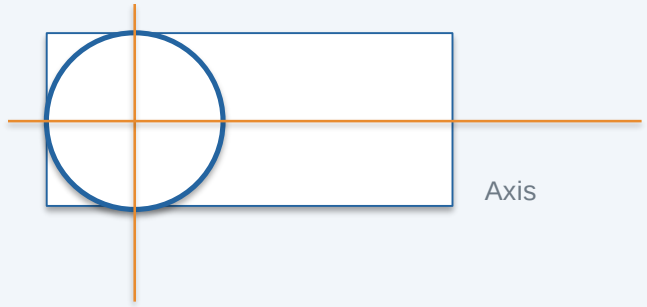
Tolerance-zone evaluation—not a single angle reading—determ

35. Worked Example: Runout

Inspection example

- A shaft journal establishes datum axis A; a second cylindrical surface has circ
- Mount the shaft using a setup that establishes datum axis A reliably.
- Rotate one full revolution and record indicator variation at a specified cross
- Total indicator reading (TIR) is the maximum minus minimum reading for th
- Acceptance depends on the specified runout control and proper setup; TIR GD&T

SCHEMATIC



KEY POINT

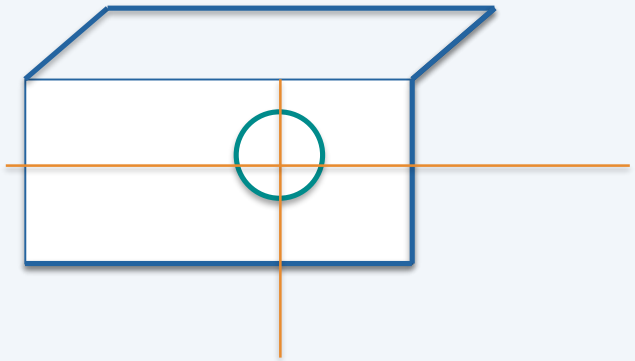
The datum axis and setup are part of the requirement.

36. GD&T vs. Traditional ± Dimensioning

Comparison

- ± dimensions are intuitive for size and coordinate limits but can overconstrain
- GD&T defines functional tolerance zones and datum relationships.
- Position tolerances can better represent bolt-pattern assembly requiremen
- Profile can simplify complex contours compared with many individual coor
- Use both dimensioning methods where appropriate; GD&T is not a replace

SCHEMATIC



KEY POINT

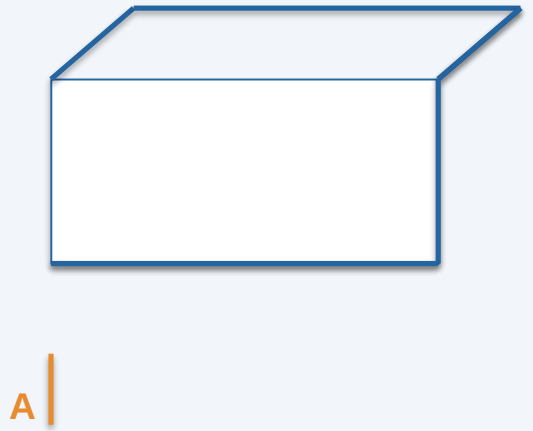
Choose the clearest, functionally correct specification.

37. Design Review Checklist

For design engineers

- Does each control protect a real functional requirement?
- Are datum features stable, accessible and related to assembly function?
- Are basic dimensions and tolerance values complete and unambiguous?
- Are modifiers and datum material-boundary requirements intentional?
- Can the supplier inspect the requirement with a realistic method?
- Are tolerances achievable for the process, cost and measurement capabili

SCHEMATIC



KEY POINT

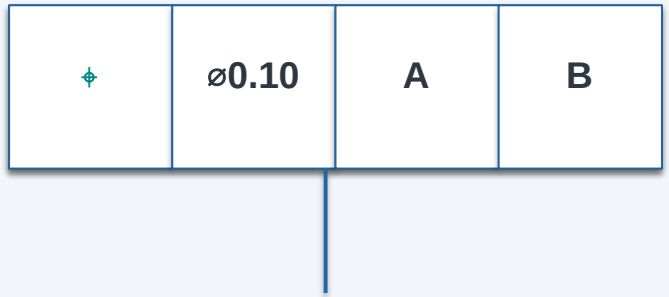
A good GD&T drawing is functional, clear and inspectable.

38. Quick Quiz

Test your understanding

- 1) Which family includes flatness and circularity?
- 2) Does flatness require a datum reference?
- 3) For a hole, what size is MMC?
- 4) What does a diameter symbol before a position tolerance commonly ind
- 5) What is the key difference between circular runout and total runout?
- 6) What should you check before interpreting a legacy drawing?

SCHEMATIC



KEY POINT

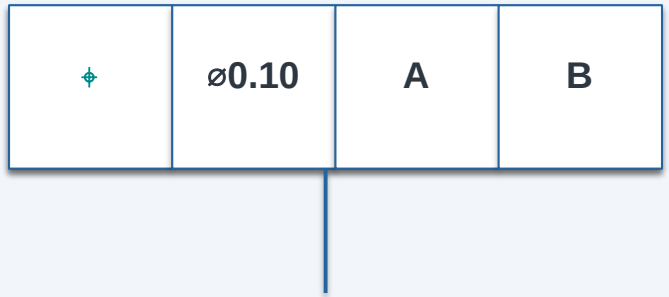
Answer key follows.

39. Quiz Answer Key

Review answers

- 1) Form.
- 2) No; flatness is a form control.
- 3) The smallest permitted hole diameter.
- 4) A cylindrical tolerance zone for the controlled axis/derived median line, v
- 5) Circular runout evaluates cross-sections; total runout evaluates the full s
- 6) The governing standard and edition, plus drawing notes and engineering

SCHEMATIC



KEY POINT

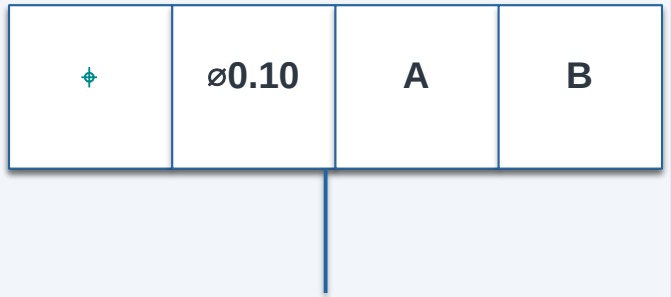
When uncertain, confirm interpretation with the drawing owner

40. Final Revision & Safety Notes

Key takeaways

- Read the complete feature control frame and related dimensions.
- Identify the controlled feature and the datum reference frame.
- Understand the tolerance zone and all modifiers before measuring.
- Use the correct standard and edition; ASME Y14.5 and ISO GPS are not i
- Use official standards for production acceptance, supplier disputes and reg
- Training deck is educational; it does not replace licensed/controlled standa

SCHEMATIC



KEY POINT

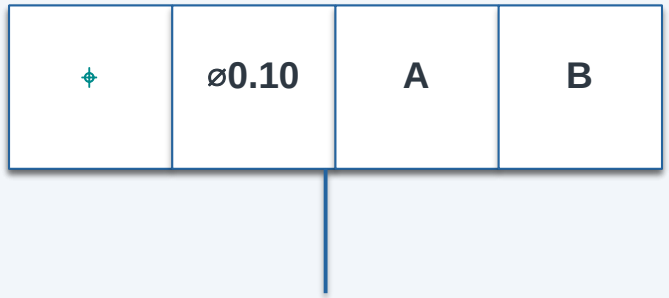
Interpretation must follow the drawing's specified standard.

41. Further Study & Reference

Use controlled, current editions for contractual interpretation

- ASME Y14.5 — Dimensioning and Tolerancing (use the edition named on the
- ISO GPS standards — relevant standards may include ISO 1101 (geometr 015 (fur
- Company drafting manual, inspection procedure, customer-specific require
- Training recommendation: practise reading real drawings, setting datums a
- Important: this presentation summarizes concepts; exact symbol rules, def d- and

REFERENCE / PRACTICE



PRACTICE

Read the frame, identify datums, sketch the zone, plan inspecti